Venkata Lakshmi Parimala Pasupuleti

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SUMMARY

Full-stack engineer with **7+ years building cloud-native data and analytics platforms** in production enterprise environments - delivering observable, resilient, and secure systems on AWS. Strong in **Python (NumPy, Pandas, scikit-learn, Flask, Boto3), React, and AWS** (Lambda, SQS, SNS, S3, Glue ETL, Step Functions, ECS, API Gateway) with hands-on experience in **serverless architecture, microservices, and CI/CD pipelines**. At Entrata (100K+ MAU SaaS), owned full-stack feature delivery end-to-end: cloud data pipelines processing millions of records, async distributed systems (RabbitMQ/SQS), and internal analytics tooling adopted across teams. At Intuit, **builds and validates next-generation cloud platform capabilities**: POC-to-production lifecycle, ML pipelines (scikit-learn, XGBoost), and data services on AWS for financial workflows. Built **CrashClarity** - an end-to-end ML analytics platform (104K records, scikit-learn, interactive dashboard) and **AgentProbe** - a Python/FastAPI platform integrating five cloud AI services. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

CORE COMPETENCIES

Python (NumPy, Pandas, scikit-learn, Flask, Boto3, Dask) | React / TypeScript / JavaScript / Node.js

AWS (Lambda, SQS, SNS, S3, EC2, ECS, Glue, Step Functions, API Gateway) | Serverless Architecture & Microservices

Data Pipelines & ETL (Glue, S3, PostgreSQL, Athena) | ML Pipelines (scikit-learn, XGBoost, SageMaker)

Docker / Kubernetes / Terraform / CI/CD | Secure & Observable API Design | Event-Driven Architecture (RabbitMQ, SQS, Kafka)

PostgreSQL / MongoDB / Redis / NoSQL | Full-Stack Feature Delivery (Design → Production) | Agile / POC Validation / Tech Enablement

EXPERIENCE

Intuit

Jun 2026 -- Present

Software Engineer - Cloud Platform & Data Services - Virginia, USA

- **Validate and deploy AWS cloud platform patterns** - POC-to-production lifecycle for Python/FastAPI microservices on AWS (Lambda, ECS, S3, API Gateway, SQS): scoping integration requirements, validating architecture proposals, and deploying via GitHub Actions CI/CD with Terraform IaC.
- Build **ML data pipelines** (scikit-learn, XGBoost): data ingestion, feature engineering, model evaluation, and prediction APIs for financial analytics workflows - delivered end-to-end from prototype through production.
- Implement **observable, resilient, and secure services**: structured logging, Prometheus/Grafana monitoring, input validation, PII handling, and fallback logic for regulated financial data environments.
- Facilitate **tech talks and upskilling** on cloud-native patterns; evaluate emerging tools and frameworks through rigorous experimentation; collaborate directly with product and platform engineering stakeholders.

Entrata - Multifamily & Affordable Housing SaaS

Oct 2020 -- Jul 2024

Senior Software Engineer - 100K+ monthly active users · Cloud data & analytics platform

- Delivered **20+ full-stack features end-to-end** - Python, React, TypeScript, Node.js, PostgreSQL, AWS - across cloud data and analytics platforms used by enterprise customers; owned design, development, testing, deployment, and production support without handoffs.
- Built **cloud data pipelines and ETL workflows** on AWS (S3, Glue, Lambda, SQS, RDS): Python/PySpark processing 1.6M+ records per run, data validation, reconciliation logic, and rollback checks; optimized 200+ SQL queries (30% performance gain).
- Designed and shipped **event-driven distributed systems**: RabbitMQ/SQS async processing with idempotency and retry logic; Docker microservices on Kubernetes (50% better scalability, 30% faster deployments) with Terraform IaC and GitHub Actions CI/CD.
- Built internal **analytics and debugging tooling** adopted across teams - Python/Bash automation eliminating 80% of manual operational work; observability dashboards reducing triage time by 40%; reviewed 300+ PRs, mentored junior engineers.

Entrata India

Sep 2018 -- Oct 2020

Junior Software Engineer - **Pune, India**

- Built Python/Java/PostgreSQL modules and data import pipelines (1.6M records); AWS S3 cloud migration; full-stack React/PHP/Node.js feature delivery.

PROJECTS

CrashClarity - End-to-End ML Analytics Platform (104K Records)

Python · scikit-learn · NumPy · Pandas · R · Dashboard · ML Pipeline

Built a full **cloud analytics platform** on 104K+ records: ingestion, cleaning, feature engineering, ML modeling (Random Forest + Logistic Regression, 85%+ accuracy), and an **interactive reporting dashboard** for non-technical stakeholders. Demonstrates end-to-end data platform delivery from raw data through production-ready ML analytics - the same pattern as QEG's signal generation and back-testing infrastructure.

github.com/venkatalakshmiparimala

AgentProbe - Multi-Provider Cloud AI Platform

Python · FastAPI · Flask · Boto3 · AWS · CI/CD · REST APIs · 5 Providers

15-endpoint FastAPI service integrating five AI cloud providers via **secure REST API design** and Boto3/AWS integrations. **Serverless-compatible architecture** with automated CI/CD regression pipelines (GitHub Actions + golden datasets). Demonstrates cloud platform engineering: observable, resilient API design, POC-to-production delivery, and multi-provider integration patterns.

github.com/venkatalakshmiparimala/AgentProbe

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia — May 2026 | GPA 3.87

B.Tech Computer Science & Engineering — JNTU Anantapur — 2017